

Macro View

Financial planning for the next quarter

67,582

B R L

Predicted 90-day revenue from the existing 75K customer base.

Actual holdout: 66,727 BRL. Model error 1.3%.

MODEL COMPARISON

XGBoost

MAE 1.65 RMSE 16.99

Total predicted: 67,582 BRL

BG/NBD + Gamma-Gamma

MAE 2.56 RMSE 18.31

Total predicted: 129,335 BRL (~2x overestimate)

Recommendation. Use XGBoost as the headline forecast. BG/NBD inflates the total because Gamma-Gamma assigns every one-shot customer (97% of the base) a positive monetary prior, which compounds across the population.

Micro View

Where to spend the marketing budget

01

Win-back 60+

high-value at-risk customers

frequency_cal ≥ 1 , p_alive < 0.5 , T_cal $\geq 90d$, ranked by CLTV

Personalised discount, category email, freight subsidy on preferred category.

02

Convert 97%

one-shot customers

frequency_cal = 0 across the observation window

Day-30 follow-up email plus 60-day coupon. Biggest CLTV lever.

03

Protect Top 5%

active high-value cohort

p_alive > 0.7 and CLTV in the top tail

Loyalty perks: early access, priority shipping. No discount erosion.

Strategic priority. The win-back universe is small because Olist is one-shot. The largest CLTV opportunity is structural: converting first-time buyers into second-time buyers. A one-percentage-point lift in repeat rate would compound into substantial incremental revenue.